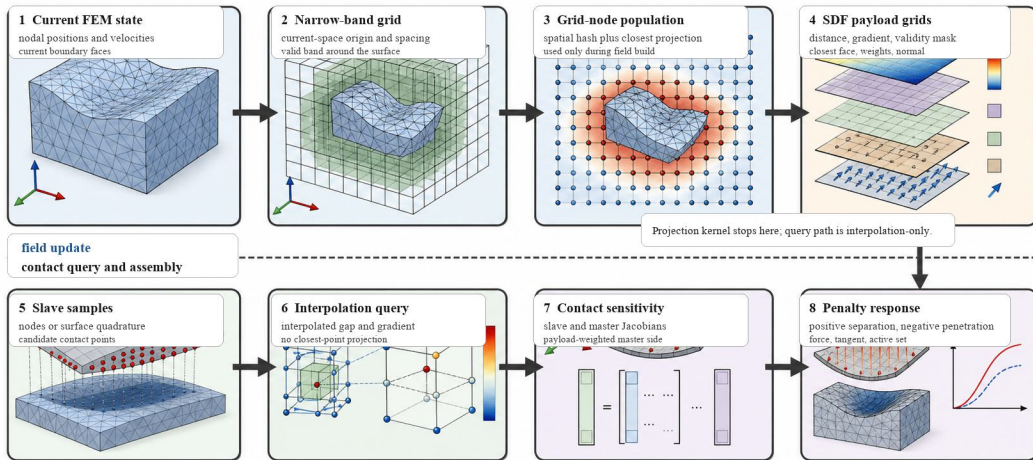


Dynamic narrow-band SDF update and field-contact query

Projection is internal to field construction; contact queries interpolate the current SDF field and payload.



Amortization gate: $T_{\text{update}} + Q T_{\text{field}} + Q_r T_{\text{refine}} < Q T_{\text{projection}}$

Report crossover Q^* ; acceleration is claimed only beyond the measured crossover.